Vermont Snowfall Analysis and Prediction Using Machine Learning Techniques

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1 Introduction

In New England and the wider Northeast area, rising winter temperatures are expected to alter the frequency of snow events, along with trends of rain replacing snow precipitation. These regional impacts of global climate change could create severe compounding effects; therefore we must analyze existing data to see the levels of change and the future consequences of our lack of environmental protection. Our project will use NOAA daily snowfall and snow depth observations to model long term snowfall trends, identify which parts of New England have been and will be most impacted, and use machine learning models to predict snowfall for the current season and future seasons at individual weather stations and averaged regional clusters.

2 Problem Definition

Snowfall is highly variable by year, so separating standard noise and natural change from human driven climate change is a modeling challenge. Prior regional studies show winter warming and snow reductions in the Northeast, so we will conduct this updated spatial analysis to determine if trends have changed. Our goal was to determine useful information about the future of winter in Vermont. The modeling approach focuses on identifying long-term trend estimates and generating future forecasts at the regional level. Key metrics include seasonal totals, season length, and the dates of first and last snowfall. By analyzing trend significance and shifts in snow-to-precipitation ratios, the study highlights which regions have experienced the most substantial environmental changes.

3 Dataset

We used the publicly available NOAA/NCEI "Daily U.S. Snowfall and Snow Depth" website to obtain daily station data for snowfall and snow depth. The data is recorded by a vast array of weather stations across the nation, however the scope of this project limits our search to those located in Vermont. Collecting the data for our purposes was a time-consuming process bottlenecked by a somewhat antiquated user interface centered around data "orders" that limited requests to about 10 stations per order. With 112 valid stations, the data collection process took more effort than expected. A dictionary is included with the data which describes each column.

NOAA states that the "GHCN (Global Historical Climatology Network)-Daily is a database that addresses the critical need for historical daily temperature, precipitation, and snow records over global land areas. GHCN-Daily is a composite of climate records from numerous sources that were merged and then subjected to a suite of quality assurance reviews. The archive includes over 40 meteorological elements including temperature daily maximum/minimum, temperature at observation time, precipitation, snowfall, snow depth, evaporation, wind movement, wind maximums, soil temperature, cloudiness, and more." (NOAA). Snowfall was requested in centimeters, although the column also includes "T" for trace amounts and "M" for missing values. Records containing either of these values were removed.

Each record also contains the latitude and longitude of the observation's stations. Records across all stations ranged from as far back as the early 2000's and were requested to December 31st, 2025. To enforce some assumptions, records were then filtered to 2009 and later. Days with missing values in key columns were removed.

4 Algorithm Definition

The workflow for this project combines machine learning regression models with temporal features, a time series split validation, a multi step forecasting, and a per region statistical trend test. We believe that tree based ensembles and linear models are better matched to this project scope, and will test the performance of these different models in forecasting.

4.1 Data Aggregation and Feature Engineering

The GHCN daily records are collapsed into one row for each winter season for every station, aggregating totals, maxima, means, and counts. Lag years and multiyear rolling average features were added based on prior seasons, computed for each station-season row. The rolling means are computed with a one-season shift so that the current target is never included in its own predictors. This shift is necessary because without that year omission, there could be target leakage and the model would produce overly optimistic scores. Seasons at the start of a station's record for which lag features are undefined are dropped. The next step was to partition Vermont regions based on latitude: the region index (1–4) enters the model as a feature and is also used in the aggregation process for region trend statistics. The latitude split for grouping stations was chosen over clustering on K nearest neighbors or elevation because it made for better defined zones that more accurately capture broad trends of climate patterns across the state.

4.2 Model Comparison

Five models were compared based on forecasting capability, with simple linear models to flexible non-linear ensembles. These models were chosen as snowfall can be driven by a high density of fairly smooth relationships with some outlying nonlinear effects:

- **Linear Regression:** Assumes snowfall is a linear function of the 19 features.
- **Ridge Regression** ($\alpha = 1.0$): Adds an L^2 penalty on coefficients to minimize large weights and stabilize estimates among the lag and rolling features.
- **Lasso Regression** ($\alpha = 1.0$): Adds an L^1 penalty which can push coefficients to zero and thereby perform feature selection.
- **Random Forest Regressor** (200 Trees, Max. Depth = 15, Min. Leaf Samples = 5): An ensemble of decision trees trained via bagging, which captures interactions without requiring feature scaling.

- **Gradient Boosting Regressor** (200 Trees, Max. Depth = 5, Learning Rate = 0.1): Unlike Random Forest's parallel bagging, boosting proceeds sequentially allowing for each new tree to correct errors on seasons the ensemble has handled poorly.

4.3 Recursive Forecasting

Projecting snowfall five years into the future involved using Random Forests for a multistep forecaster as it was found to be the best performing model. The Random Forest is retrained on all available data with no hold out, and for each station we seed the forecast with that station's most recent observed feature vector. At each horizon (1–5 years) we shift the snowfall lag window so that it becomes the new lag -1 and repeat. As the feature set also includes lagged precipitation, snow days, and snow/precipitation ratio (which our model does not predict) we estimate these quantities from the predicted snowfall using per station scaling factors computed as historical medians. Medians are used here for handling extreme years, and rolling averages are maintained through a per station buffer of the last 3 or 5 (observed or predicted) season snowfall values so that `snow_roll3` and `snow_roll5` remain true means.

4.4 Trend Analysis

Prior to evaluating the model, the full station-season dataset was assessed to identify any obvious trends before seeing any model results via two tests. Ordinary least squares (OLS) linear regression between the year and mean seasonal snowfall yields a slope in mm/year and an associated p-value that represents the significance of the trend. The Mann-Kendall Tau test checks whether snowfall trends in a direction without any assumption of linearity. Agreement between the two tests strengthens claims of directional change, whereas conflicting results mean that the trends are too sensitive to modeling assumptions.

Spatial and temporal factors were considered in the analysis when selecting models to predict snowfall. First, a heuristic approach is used to separate stations based on latitude. Stations are assigned to a region R based on a vector of latitude thresholds $L = [43.50, 44.00, 44.50]$. This segments the state into four distinct meteorological zones:

- **Southern VT:** $\text{lat} < 43.50$
- **Central VT:** $43.50 \leq \text{lat} < 44.00$
- **Northern VT:** $44.00 \leq \text{lat} < 44.50$
- **NEK/Border:** $\text{lat} \geq 44.50$

To enable the models to capture multi-year snowfall trends, a temporal windowing function is implemented. For a given station at a given winter season, lagged observations were generated, providing the model with the snowfall totals from the three preceding seasons. Moving averages of both 3 and 5 years are also used to remove noise and create more stable predictors.

Prediction is evaluated using five distinct algorithms to compare their ability to handle the relationship between elevation and snowfall. The baseline is normal linear regression, with improvements and changes in methodology being made incrementally. Because of the high dimension of the feature set (19), regularization is used for both Ridge Regression and Lasso Regression.

Another forecasting model employs a multi-step approach to project Vermont's seasonal snowfall five years into the future. By utilizing a random forest regressor with 200 predictors trained on historical data dating back to 2011, the system analyzes a complex feature set including geographic metadata (latitude, longitude, and elevation) and lagged snowfall totals and rolling averages. In this framework, the model predicts the immediate next season ($T + 1$), then feeds that prediction back into the feature vector as a "known" lag to generate the subsequent year's forecast ($T + 2$). This iterative process allows the model to capture potential medium-term climatic shifts, though it relies on the stability of historical patterns to ground its future projections.

The Gradient Boosting Regressor is an additive ensemble method used in this pipeline to capture complex, non-linear relationships between geographic features and seasonal snowfall. Unlike the Random Forest, which builds independent trees in parallel, Gradient Boosting constructs decision trees sequentially, with each new tree specifically trained to correct the residual errors of the combined ensemble that preceded it. The model uses 200 estimators with a learning rate of 0.1, allowing it to fine-tune its predictions for difficult-to-forecast seasons.

5 Methodology

5.1 Evaluation Criteria

Model performance is measured by training on the statewide station-season training set, then testing on the most recent 3 held-out seasons, and evaluating the resulting three metrics:

- **Root Mean Squared Error (RMSE)** — The primary metric for model selection reported in millimeters, RMSE penalizes large errors more heavily than small ones which is necessary for our dataset. Mispredicting an extreme snow season by a wide margin is more costly than a small error in a typical season
- **Mean Absolute Error (MAE)** — Reported in millimeters alongside RMSE, MAE is more robust to outlier seasons and gives a more typical error magnitude. The ratio RMSE/MAE diagnoses whether errors are concentrated in a few large misses (high ratio) or distributed uniformly (ratio near 1)
- **Coefficient of determination (R^2)** — The proportion of variance in total seasonal snowfall explained by the model. A value of 1 indicates perfect prediction, while a value of 0 indicates the model does no better than just predicting the mean

These metrics help determine which model is best at prediction by picking whichever has the lowest RMSE and MAE, along with the highest R^2. We used this model to then retrain on the full statewide

station-season dataset (not holding out recent years) and use it to predict the snowfall for the next 5 seasons at every station.

5.2 Hypotheses Testing

The experimental design is structured around two hypotheses. The first is that non-linear ensemble models (Random Forest, Gradient Boosting) will outperform linear models (Linear, Ridge, Lasso) because the causal relationship between snowfall and its effects is not linear. The second is that there exists a general trend of decline across Vermont for seasonal snowfall totals.

5.3 Training and Test Split

After cleaning and engineering, the dataset consists of daily GHCN station observations for Vermont from 2009 through early 2026, filtered to stations that report at least one valid snowfall observation and to station-seasons with at least 30 days of snowfall observations. After aggregation to one row per station per winter season, with lag years and rolling mean year features attached, the dataset was recorded as `ml_ready`. The split between training and test sets is only time based; the 3 most recent complete winter seasons form the test set and all earlier seasons form the training set. This is the split that simulates the forecasting problem by predicting conditions that have not yet been observed. It is realistic in forcing the model to extrapolate along the time axis, not merely interpolate between nearby stations in already observed years.

5.4 Experimental Methodology

Before evaluating on the held-out test years we run `sklearn.TimeSeriesSplit()` with five folds on the training set. This function produces expanding window folds in which each fold trains on all seasons up to a cutoff and then tests on the next grouping of seasons. This preserves the time ordering and doesn't allow the model to see future data during training. This gives a distribution of RMSE values across five forecast scenarios. Reporting CV mean and standard deviation of RMSE allows us to check that a single test set score is not a result of one abnormally easy or hard 3 year window.

Next, each model is retrained on the full training set and evaluated once on the three held-out seasons. Linear models are trained on standardized features, whereas tree ensembles use raw features. The model with the lowest test RMSE is set as the best model and is used for future forecasting.

5.5 Presentation and Findings

Regional means are computed by averaging snow totals across all stations within each latitude band per season. Each region's time series is tested for trend using OLS linear regression (slope in mm/year, p-value, R^2) and Mann-Kendall Tau (rank correlation and p value) with $\alpha = 0.05$. For the five regression models, the held-out test RMSE, MAE, and R^2, as well as the cross validated mean and standard

deviation of RMSE from `TimeSeriesSplit` are observed. The best model is further visualized through a predicted-vs-actual scatter (with a $y = x$ reference line) and a histogram of residuals so that both global fit and error distribution are visible.

For the trend analysis, four small-multiple panels (one per region) are created, each showing the annual regional-mean snowfall as bars, the fitted OLS line as a dashed overlay, and the slope and p-value in the legend, so that statistical and visual evidence appear together.

As for the forecaster, the predicted mean snowfall by region is reported for each of the next five seasons, overlaid on historical bars with error bars showing the standard error of the mean across stations within each region, and a dashed line at the historical per-region mean for reference.

6 Discussion

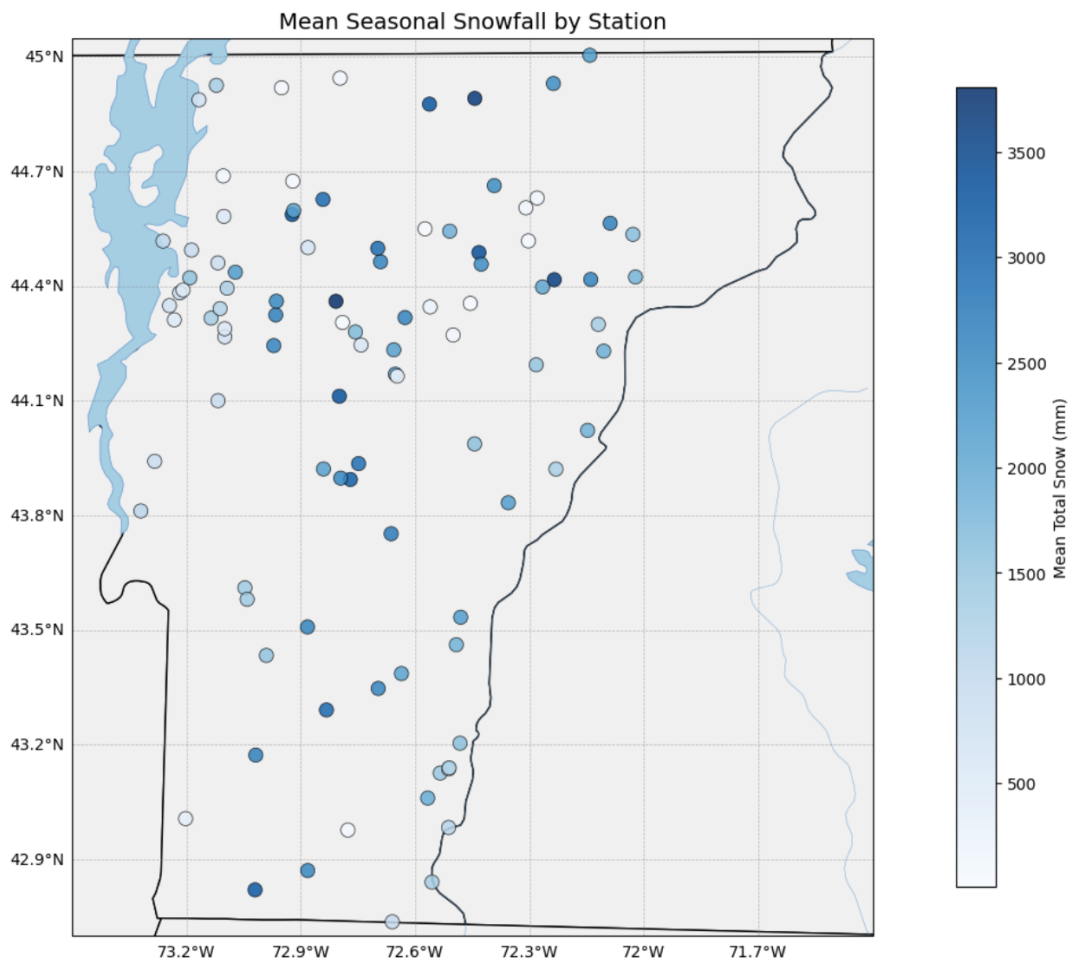

Figure 1: Mean Seasonal Snowfall by Station: *This map reveals the disparity in the number of stations across the different latitudes in Vermont. The stations are colored by average reported snowfall depth. The substantially higher density of stations in the North and Central VT regions can be observed*

Region	Latitude Range	Number of Stations	OLS Slope (mm/year)	R Squared	Kendall Tau
NEK/Border	44.50 <= 45.00	14	-41.6	0.109	-0.183
Northern VT	44.00 <= 44.50	24	-10.7	0.014	-0.100
Central VT	43.50 <= 44.00	41	33.7	0.067	0.029
Southern VT	< 43.50	18	-59.7	0.220	-0.486

Figure 2: OLS and R^2 Analysis on Raw Dataset: *This table displays figures that clarify the physical differences in regions, both in number of stations and latitude, while also showing snowfall trend data. With respect to a p-value of 0.05, none of the tests found significance.*

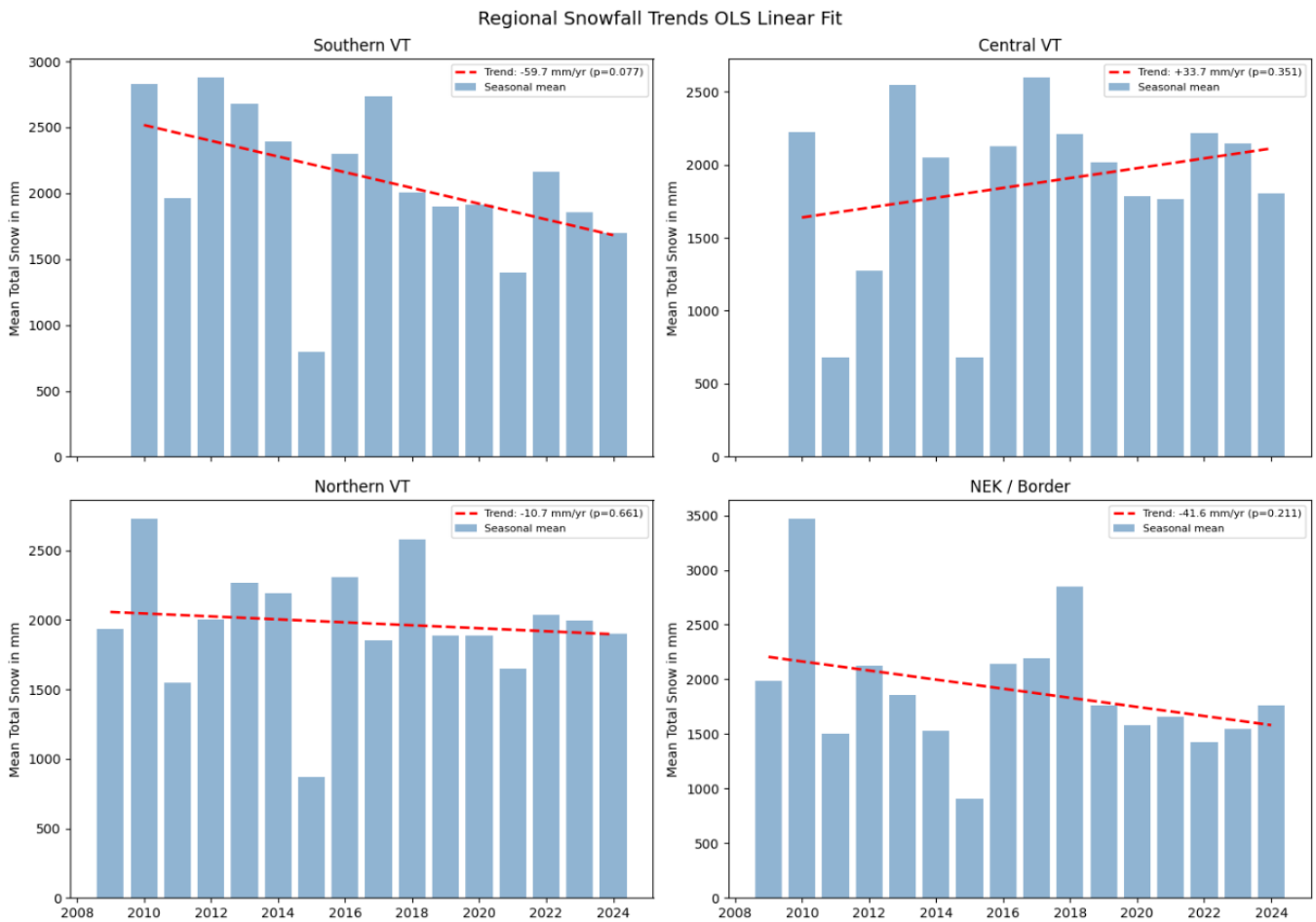

Figure 3: Regional Snowfall Trends OLS (Linear Fit): *These graphs emphasize the differences in snowfall trends across different regions in Vermont. Southern, central, and northern Vermont seems to be showing a negative trend, while central Vermont stands out for being the only region to have a positive trend.*

Model Type	RMSE	MAE	R Squared
Random Forest	634.428645	481.808733	0.698985
Linear Regression	646.966510	499.347463	0.686970
Lasso Regression	647.261909	499.795156	0.686684
Ridge Regression	648.910413	500.712075	0.685086
Gradient Boosting	706.786240	536.613028	0.626407

Figure 4: Results by Model: *This table shows model types with their RMSE, MAE, R^2. Based on this information, the random forest seems to have the least errors, so we went through with using Random Forest for predicting the next 5 years.*

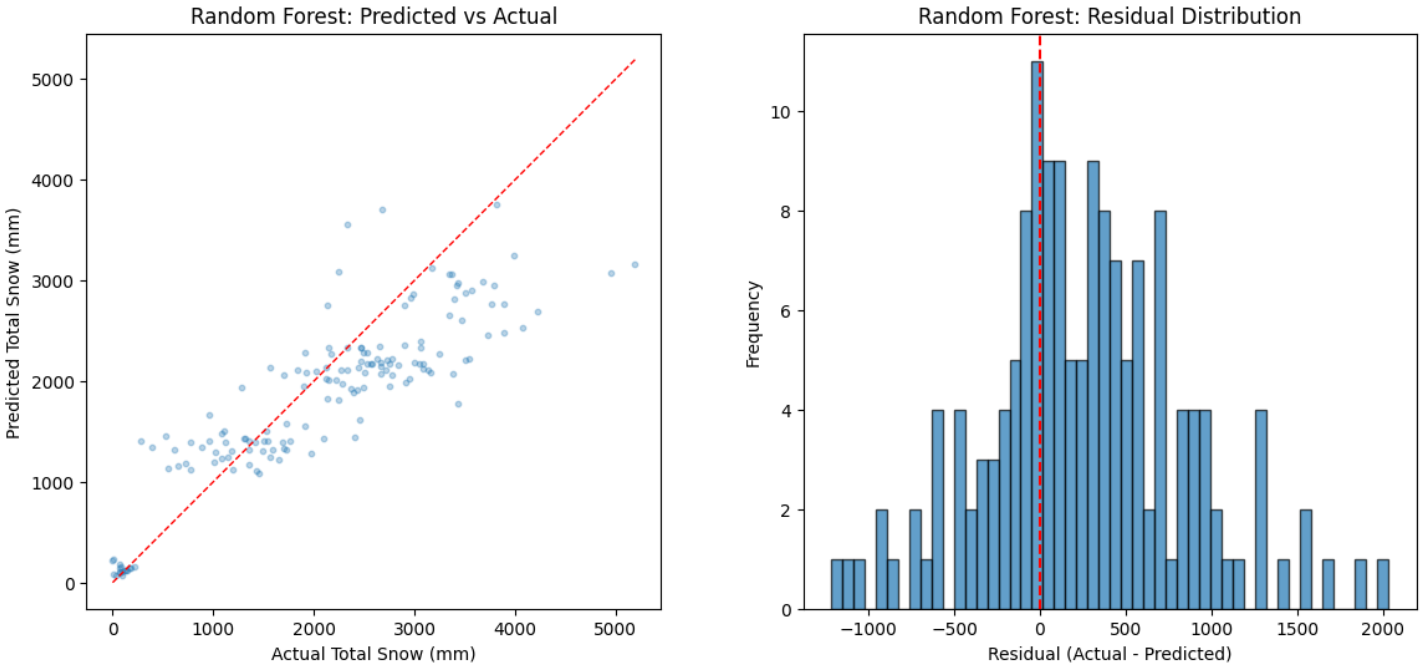

Figure 5: Random Forest Residuals: *The residuals of the random forest predictions are shown, with more positive residuals than negative, suggesting the model tends to underestimate snow depth numbers*

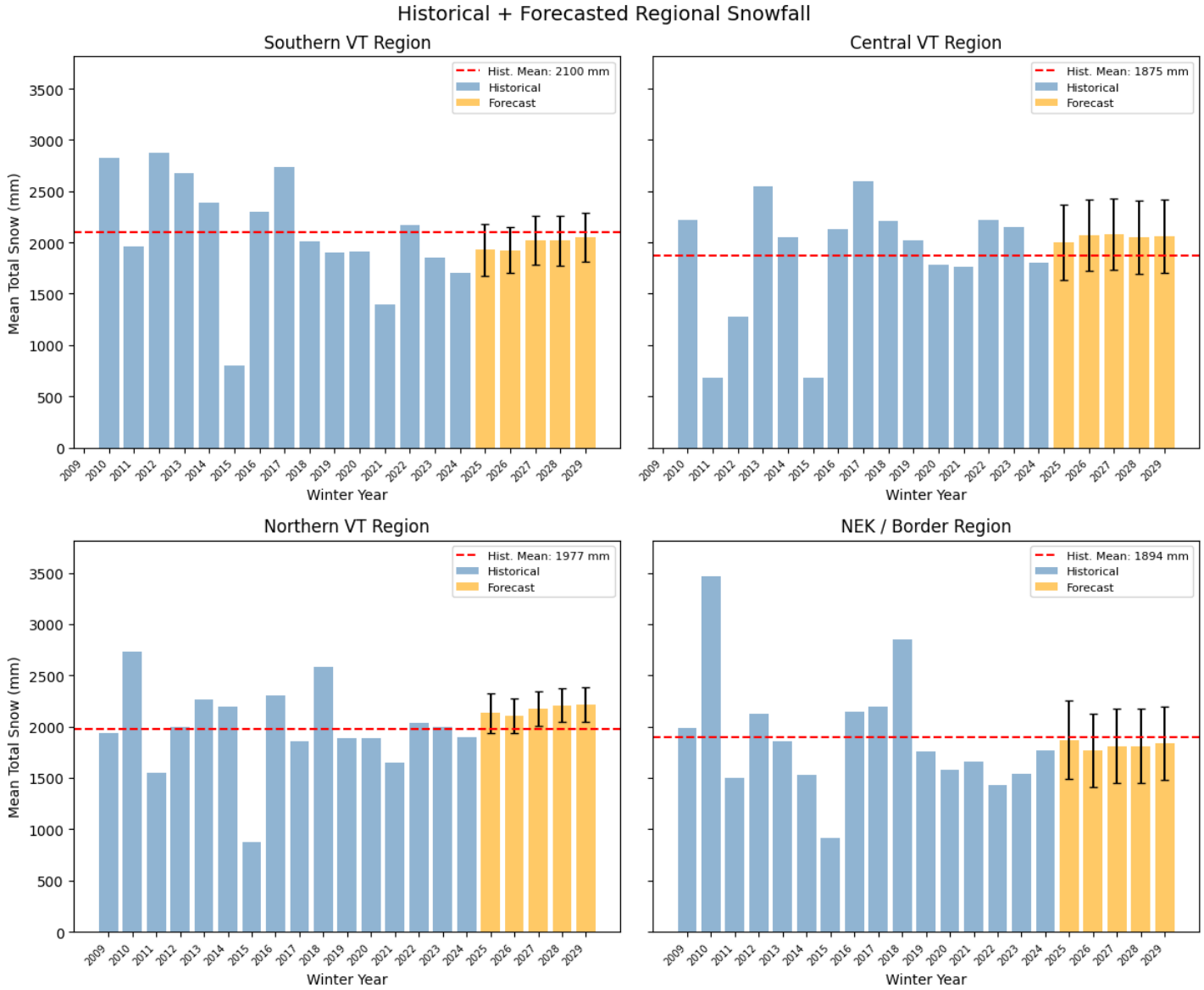

Figure 6: Forecasted Snowfall by Region: *These graphs show 5 years of predicted snowfall totals by region, starting in 2025. The error bars differ in length, displaying prediction variance across the different regions. The mean bar shows the differences in predicted snowfall data to the historical average*

Based on observed climate conditions we hypothesized that the model would predict a yearly decrease in snowfall depth, but the random forest model (Figure 4, 5, 6) yielded mixed results. The southern and NEK regions are expected to have lower than average snowfall each of the next 5 years. The other two regions, central and northern, are expected to have snowfall above the historical average in all of the following 5 years. In the context of our expectation of statewide snowfall decrease, it's especially surprising to look at the residual graphs (Figure 5) and see a considerable amount of the residuals be positive. This indicates that not only are the existing snowfall predictions higher than expected, but they are tending to underestimate the actual snowfall numbers.

These trends differ slightly from the trendlines found in the OLS graphs (Figure 2, 3) with the northern region having a negative linear trend in the OLS model and above the mean predictions in the Random Forest model. These differences can be attributed to the difference in the type of models: Random Forest accounts for outliers, interaction, and nonlinearity for the dataset to make predictions, whereas OLS accounts for a linear analysis of the raw dataset without complexity. The OLS model's simplicity, with its only variable being year, can be seen to not find any significant trend among the data, with R^2 values ranging 0.014 to 0.220 (Figure 2). The Kendall Tau results are similarly insignificant in determining a trend for each region, signifying a lack of any consistent identifiable trend between snowfall and time. These tests suggest that this is not a dataset with a simple trend for snowfall, and that a ML model is necessary to make informed forecast predictions.

We then analyzed the performance of the five models, with R^2 values ranging from 0.623 to 0.699 (Figure 4). This small range displays the limited effect different models have on representing our dataset; the regression models are especially close with all three models within 0.002 of each other. This demonstrates the limited effect regularization has on the data, signifying that the model doesn't tend to naturally overfit and lacks collinearity among features.

The error bars (Figure 6-) indicate a lack in confidence of these previously indicated trends as many intervals include the area mean, suggesting that it's not unlikely for the snowfall totals to be on the other side of the historical average. While it is worth mentioning that some of the regions have smaller error bars (northern and southern) the model's overall R^2 value of approximately 0.699 (Figure 4, 6) reflects the lack of prediction accuracy. This lack of confidence in predictions prevents making meaningful conclusions to our hypothesis.

This limited accuracy of even the best performing model sheds light on issues with the dataset as a whole. While the dataset was rather large, the number of usable seasons was relatively small. Many stations went without the funding and/or technology necessary to develop consistent data collection methods until 2009. The lack of power across several models demonstrates that having less than 20 years of data is not enough to establish meaningful results. Additionally, there is a lack of consistency in data collection methods among the different weather stations, which could lead to unnatural discrepancies between stations.

7 Related Work

7.1 Estimation of Frequency Based Snowfall Depth Considering Climate Change Using Neural Network

The study predicts future extreme snowfall in Korea using climate model data and nonlinear models (neural networks and regression) based on temperature and precipitation. It finds that frequency-based extreme snowfall depths decrease over time under climate change. Our approach could differ by using a different prediction model or reversing the explanatory and response variables.

7.2 Predicting Snow-to-Liquid Ratio in the Mountains of the Western United States

The study uses machine learning methods to better predict snow forecasts. It focuses a lot on the relationship snowfall has with factors like snow-to-liquid ratio (SLR) and snow-water-equivalent (SWE). They used features like temperature, wind speed, humidity, and others to predict SLR. Utilizing a random forest yielded them a superior model to previously existing ones, with nearly half of the variability represented, much better than the previous less than 25% variability represented. In our project, we could try analyzing how the median SLR has been changing over time.

7.3 Changes in the proportion of precipitation occurring as snow in New England

The study uses snow / precipitation ratio collected from multiple stations as both individual and aggregate sites, creating time series. These were analyzed with a "nonparametric test for monotonic trend based on Kendall's tau statistic", and the sites with significant p-values were determined to create a trend of decreasing S/P. Our methods differ in using machine learning methods like random forest and linear regression compared to this study's time series statistics analysis.

8 Code and Dataset

Code was written using Google CoLab and hosted on GitHub. Instructions for replicating/running this project:

1. Navigate to the public GitHub page listed below. Download the zip file including the `project.ipynb`.
2. Ensure that you have the necessary pip packages installed, listed within the first section of the Jupyter notebook.
3. Move the csv labeled "full" from the zip archive to the same working directory as the `project.ipynb`.

Each Jupyter cell should be run sequentially, or alternatively, the "Run All" feature can be used. The dataset must be reloaded for sections to rerun.

https://github.com/TedPittman/SnowfallForecast

9 Conclusion

This study analyzed long-term snowfall trends across Vermont using NOAA GHCN-Daily snow data and applied multiple machine learning models to forecast seasonal snowfall. The results indicate a generally decreasing but statistically insignificant trend in seasonal snowfall across the state, with southern Vermont stations showing the most pronounced reduction in snowfall totals. While snowfall predictions can be made, the lack of statistical power of the forecasts highlights the importance of having a sufficient amount of consistent, high-quality data for temporal analysis. To truly look at the effect of climate change, it is necessary to gather data from when the climate has substantial differences as such types of data is better for explaining long term change in snowfall trends. Given the current amount of data spanning fewer than two decades and a lack of data quality, the variance within predictions prevents meaningful conclusions to the hypothesis. Additional data that goes further back in time would likely decrease the prediction variance and generate a more accurate historical mean

References

[1] Burakowski, E. A., Wake, C. P., Braswell, B., & Brown, D. P. (2008). Trends in wintertime climate in the northeastern United States: 1965–2005. *Journal of Geophysical Research: Atmospheres*. https://doi.org/10.1029/2008JD009870

[2] Veals, P. G., Pletcher, M., Schwartz, A. J., Chase, R. J., Harnos, K., Correia, J., Wessler, M. E., & Steenburgh, W. J. (2025). Predicting Snow-to-Liquid Ratio in the Mountains of the Western United States. *Weather and Forecasting*, 40(10), 1825–1838. https://doi.org/10.1175/WAF-D-24-0233.1

[3] Durre, I., Menne, M. J., Gleason, B. E., Houston, T. G., & Vose, R. S. (2010). Comprehensive automated quality assurance of daily surface observations. *Journal of Applied Meteorology and Climatology*, 49(8), 1615–1633.

[4] Huntington, T. G. (2004). Changes in the proportion of precipitation occurring as snow in New England (1949–2000). *Journal of Climate*, 17(13), 2626–2636.

[5] Kim, Y., Kim, S., Kang, N., Kim, T., & Kim, H. (2014). Estimation of frequency based snowfall depth considering climate change using neural network. *Journal of the Korean Society of Hazard Mitigation*, 14(1), 93–108. https://doi.org/10.9798/KOSHAM.2014.14.1.93

[6] McCray, C. D., et al. (2023). Changing nature of high-impact snowfall events in eastern North America. *Journal of Geophysical Research: Atmospheres*. https://doi.org/10.1029/2023JD038804

[7] Menne, M. J., Durre, I., Vose, R. S., Gleason, B. E., & Houston, T. G. (2012). An overview of the Global Historical Climatology Network-Daily Database. *Journal of Atmospheric and Oceanic Technology*, 29(7), 897–910.

[8] NOAA/NCEI. Daily U.S. Snowfall and Snow Depth. `https://www.ncei.noaa.gov/access/monitoring/daily-snow/`

[9] NOAA/NCEI. Global Historical Climatology Network daily (GHCNd). `https://www.ncei.noaa.gov/products/land-based-station/global-historical-climatology-network-daily`